HORN: Directional Outlier Propagation in Hybrid Reasoners

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. HORN tests whether attention-to-SSM and SSM-to-attention boundaries differ in outlier magnitude and quantization-noise sensitivity. No positive result or native systems claim is made yet.

1 Current Gate

The branch must pass activation-magnitude, quantization-noise propagation, and cross-model/control gates before GPU validation.

2 Mac Artifact Status

A deterministic synthetic H1 packet validates the directional-boundary readout. It reports SSM-to-attention over attention-to-SSM max ratio 3.775 and kurtosis ratio 7.139, producing `SYNTHETIC_PASS_REAL_BOUNDARY_DUMPS_NEXT`. This is synthetic-only: it is not model evidence, not a quantization-noise result, and not GPU evidence.

3 Next Evidence

The next admissible row is a real boundary activation dump from a hybrid model. HORN remains a control branch unless real H1/H2/H3 gates show consistent directional asymmetry that pure-attention and pure-Mamba controls do not share.

4 Admissible Real Packet

The real H1 packet must use the shared config-derived architecture maps rather than substring-only module classification. Each row must report `model_id`, adjacent layer IDs, `direction`, `boundary_index`, normalization placement, `max_abs`, `rms`, `kurtosis`, and `control_type`. The required controls are boundary rows, non-boundary adjacent rows, and permuted-direction rows. The packet must include `architecture_map_hash` and pass `experimental.shared.check_gate_packet --mode real --project horn`.

Required control	Reviewer objection addressed
Explicit boundary map	Avoids substring artifacts.
Non-boundary adjacent pair	Separates boundary effects from depth effects.
Permuted direction label	Tests whether direction carries signal.
Matched normalization side	Controls RMSNorm/LayerNorm placement.
Pure-architecture control	Prevents generic outlier claims.

Table 1: Controls required before HORN can claim directional hybrid asymmetry.